

ω Centauri: first dynamical experiments

Compact dark-matter remnants and extended progenitors

oCen_dm project notes

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Abstract

We analyse the first completed batch of 24 live and frozen-potential evolutions, designed to establish numerical behaviour and guide two independent research programmes. Compact cusped and cored dark-matter distributions around a rigid nucleus complete one Galactic passage with a pericentre of 2.00 kpc. Relative to matched isolation controls, their enclosed dark-matter masses fall by 1.23% and 3.39% inside 20 pc, and by 5.35% and 7.12% inside 70 pc. The strongest depletion follows the second disc crossing. Frozen controls reproduce most of the response, while the outer velocity distribution becomes tangential. Short isolation controls show weak sensitivity of the 20-pc mass to timestep and softening, although numerical convergence of the full tidal passage remains to be tested. Extended progenitors complete 100 Myr in isolation with live and frozen enclosed masses agreeing within 1% at the sampled radii from 100 pc to 10 kpc. Their cored halo has only 16 initial particles inside 20 pc, which prevents a useful measurement of small central changes. These pilots support the next controlled experiments: full-passage numerical comparisons for the compact remnants, and disruption of the extended progenitors with improved sampling for central-remnant studies. The analytical comparisons recover the trend of stronger outer evolution, but instantaneous energy truncation overestimates the first-passage density loss, while the scalar pericentre-shock estimate underestimates full-orbit heating. Initial radius alone does not describe the exposure of eccentric internal orbits to the Galactic tide.

1 Questions and scope

The two programmes address different questions. The *remnant-survival* experiments ask how a compact dark-matter distribution responds to Galactic tides once it surrounds an existing nucleus. The *progenitor-disruption* experiments begin with a much larger stellar body and halo, and will measure what remains after their disruption. Both are useful independently; neither requires the other programme to finish first.

This batch is the first modelling step for these questions. It tests the initial equilibria, distinguishes tidal response from evolution in isolation, and identifies useful resolution choices. It does not yet reconstruct the historical progenitor or determine a final dark-matter constraint. The nucleus is rigid and follows a prescribed track; the runs include neither dynamical friction nor a response of the nucleus to the live particles. The anisotropy specified for a simulated DM component is separate from the stellar anisotropy being fitted in the kinematic modelling sequence.

2 Models and initial conditions

2.1 Density profiles and the nucleus

Every sampled component has the spherical density law

$$\rho(r) = \rho_0 \left(\frac{r}{a}\right)^{-\gamma} \left(1 + \frac{r}{a}\right)^{\gamma-b} \exp\left[-\left(\frac{r}{r_{\text{cut}}}\right)^2\right]. \quad (1)$$

Here a is the scale radius, b the outer power-law index before the taper, and $\gamma = 1$ or 0 defines the cusp or core. The taper acts on the density; it is not an energy truncation or a boundary at an

instantaneous tidal radius. The quoted total masses are finite component masses, not cosmological virial masses. A core in this family is not a Plummer sphere.

The nucleus is an analytic Plummer potential,

$$\Phi_n(r) = -\frac{GM_n}{\sqrt{r^2 + a_n^2}}, \quad M_n = 3.55 \times 10^6 M_\odot, \quad a_n = 5.4 \text{ pc}. \quad (2)$$

It is included once in the combined satellite potential and is not represented by a softened particle. The present experiments adopt this nucleus as a fixed specification; they do not import a posterior draw from the kinematic fits.

Table 1: Continuum component specifications. The remnant cusp and core are matched at $M_{\text{DM}}(< 20 \text{ pc}) = 10^5 M_\odot$; the progenitor pair is matched in total component mass. The stellar-body parameters apply to both progenitors.

Component	γ	a [pc]	r_{cut} [pc]	b	β
Remnant DM cusp	1	20	100	3	0
Remnant DM core	0	20	100	3	0
Progenitor DM cusp	1	1000	10000	3	-0.3
Progenitor DM core	0	1000	10000	3	-0.3
Progenitor stellar body	1	500	3000	4	0

The inner-mass convention gives total remnant DM masses of $4.32466 \times 10^5 M_\odot$ for the cusp and $7.71985 \times 10^5 M_\odot$ for the core. Thus the remnant comparison changes both central slope and the mass distribution outside the matched aperture. Each progenitor instead has $10^9 M_\odot$ in DM and $10^8 M_\odot$ in its extended stellar body. Their central DM masses differ substantially (Section 8).

2.2 Distribution functions and particle sampling

The initial distribution function (DF) of each component has constant anisotropy in the potential of the nucleus plus *all* sampled components:

$$f(\mathcal{E}, L) = L^{-2\beta} f_{\mathcal{E}}(\mathcal{E}), \quad \beta = 1 - \frac{\sigma_\theta^2 + \sigma_\phi^2}{2\sigma_r^2}. \quad (3)$$

Preparation performs an independent signed inversion to detect negative DF contributions that would be hidden by clipping. The rejection threshold is a negative contribution exceeding 0.5% of the absolute inversion integral. A separate grid checks density recovery within 5% and anisotropy recovery within 0.03. In this batch, the measured maximum fractional density error is less than 1.1×10^{-4} and the maximum absolute anisotropy error is less than 3.8×10^{-4} . These checks apply to the sampled energy and radial domains; they are numerical validations rather than positivity proofs over all phase space.

The broad isotropic progenitor core fails the signed DF check in this particular combined potential. Both progenitor halos therefore use $\beta_{\text{DM}} = -0.3$, which passes; their stellar components have $\beta_\star = 0$. This choice does not assert that every cored halo requires tangential anisotropy.

Remnants use equal-mass particles. Progenitors use pericentre-based importance sampling in the initial combined spherical potential. If r_p denotes the pericentre of a particle orbit, the sampling probability factor is

$$p(r_p) = \max \left[p_{\min}, \min \left(1, \frac{r_{\text{protect}}^2}{r_p^2} \right) \right], \quad r_{\text{protect}} = 100 \text{ pc}, \quad p_{\min} = 10^{-3}. \quad (4)$$

The DF is sampled with this factor and particle weights are divided by p and normalised to the specified component mass. The weights remain fixed in time. Sampling by pericentre protects centre-crossing orbits even if their initial positions lie far from the nucleus. Softening scales as $m^{1/3}$

within each component. The minimum softenings are 2 pc for progenitor DM and 1 pc for progenitor stars.

Each baseline remnant contains 10^5 DM particles, with softening 0.2 pc and particle mass $4.325M_\odot$ or $7.720M_\odot$ for the cusp or core. Each progenitor has 2×10^5 DM and 10^5 stellar particles. The progenitor DM particle mass ranges are approximately $283\text{--}2.83 \times 10^5 M_\odot$ for the cusp and $134\text{--}1.34 \times 10^5 M_\odot$ for the core. Their global effective DM particle numbers are only 1.07×10^4 and 1.77×10^4 , respectively. Local effective counts, rather than nominal particle counts alone, determine the precision of the central diagnostics.

3 Evolution, controls and measurements

3.1 Live and frozen calculations

Live integrations use `gyrfalcON` particle self-gravity with the P1 softening kernel and the analytic moving nucleus. Frozen controls use `AGAMA` trajectories in the fixed initial satellite potential, translated along the same nucleus track. They retain identical initial particles and diagnostic masses but do not update the satellite potential in response to particle motion. Live/frozen differences therefore contain both a physical response of the self-potential and finite-particle, softening and integration effects.

The natural units are kpc, km s^{-1} and $M_\odot$, with

$$G = 4.30091727067736 \times 10^{-6} \text{ kpc (km s}^{-1}\text{)}^2 M_\odot^{-1}, \quad t_{\text{unit}} = 977.7922216807892 \text{ Myr.} \quad (5)$$

The baseline maximum block step is 0.01491993 Myr and the minimum is 0.00186499 Myr, using four levels. The acceleration-based timestep factor is `fea=0.2`, corresponding to the criterion $\tau = 0.2\sqrt{\epsilon/|\mathbf{a}|}$ before block-step assignment, and the tree opening parameter is 0.5. The timestep controls halve both maximum and minimum steps. Frozen integrations use adaptive integration with the configured accuracy 10^{-8} . Requested endpoints are rounded up to the live timestep grid, and both engines use the same resulting interval.

All preparations use seed 42. Increasing particle number creates a different sample, so a doubled- N endpoint cannot be treated as the same-realisation counterpart of the baseline. Its own frozen control is needed.

Table 2: Completed batch. Each row contains both cusp and core. “Live/frozen” means two engines per profile. The repaired softening evolution replaces the failed cusp attempt in the completed count.

Experiment	Engines	Duration [Myr]	Evolutions
Remnant baseline isolation	Live/frozen	20.00762	4
Remnant full-interval isolation	Live/frozen	88.10216	4
Remnant Galactic passage	Live/frozen	88.10216	4
Remnant half-timestep isolation	Live	20.00762	2
Remnant half-softening isolation	Live	20.00762	2
Remnant doubled- N isolation	Live/frozen	20.00762	4
Progenitor isolation	Live/frozen	100.00826	4
Total			24

The remnant output cadence is 0.5 Myr in the short controls and 0.25 Myr in the full-interval comparisons. Progenitor outputs are spaced by 2 Myr. The continuum DFs use unsoftened gravity, so isolation tests explicitly check the response to particle sampling and softened live gravity.

3.2 Orbit and initial tidal state

The tidal pair uses the saved static `McMillan17` host asset and the nearest past apocentre of the nominal present-day orbit. The initial and final apocentre radii are 6.95887 and 6.91760 kpc. The

common interval contains one pericentre, at 1.99817 kpc and elapsed time 45.11369 Myr. These are the extrema of this particular saved passage, not a minimum over a many-period orbit integration.

The Cartesian track is stored in double precision and interpolated with positions and velocities. Its maximum interpolation acceleration error relative to the saved host force is 4.01×10^{-6} . Within each profile, the passage and isolation cases have identical local initial positions, velocities, IDs and masses; the cusp and core share byte-identical orbit and host files. They start as isolated spherical equilibria inserted at apocentre, without a prior tidal-relaxation stage or adiabatic host turn-on.

3.3 Apertures, energy and binding diagnostics

For each component, the stored apertures are 3, 10, 20, 35, 50 and 70 pc. The enclosed mass and effective particle count are

$$M(< r) = \sum_{r_i \leq r} m_i, \quad N_{\text{eff}} = \frac{(\sum_i m_i)^2}{\sum_i m_i^2}. \quad (6)$$

Shell densities use $r \exp(-0.1) \leq r_i < r \exp(0.1)$, with the exact shell volume. Velocity moments are mass-weighted about their shell means and are reported with N_{eff} ; moments are undefined below three effective particles. Ratios in the remnant tidal comparisons are

$$\delta_X(t) = 100 \left[\frac{X_{\text{tidal}}(t)}{X_{\text{isolation}}(t)} - 1 \right]. \quad (7)$$

When output grids differ slightly, the comparison interpolates the isolation series to the tidal times without extrapolation. The final-10-Myr medians and 16th–84th percentiles describe temporal variation in a single realisation. They are not confidence intervals or independent repeat runs.

The binding proxy iteratively removes particles with positive energy in a self-excluded spherical particle potential plus the nucleus, measured relative to the nucleus position and velocity. It is not an exact escape classification or a conserved Jacobi energy. All particles remain in the simulation.

The host tidal tensor is $T_{ij} = \partial a_{\text{host},i} / \partial x_j$. The instantaneous tidal scale uses the largest eigenvalue of

$$\mathbf{T}_{\text{eff}} = \mathbf{T} + \omega^2 \mathbf{I} - \omega \omega^T, \quad \omega = \frac{\mathbf{R} \times \mathbf{V}}{R^2}, \quad \frac{GM_{\text{bound+nucleus}}(< r_t)}{r_t^3} = \lambda_{\text{max}}(\mathbf{T}_{\text{eff}}). \quad (8)$$

The implementation selects the outermost crossing when one exists. This spatial diagnostic excludes Euler and velocity-dependent Coriolis terms; r_t is therefore a defined scale rather than an escape surface.

Reference-energy changes use

$$E_{\text{ref}}(t) = \frac{1}{2} |\mathbf{v} - \mathbf{V}|^2 + \Phi_{\text{sat},0}(|\mathbf{x} - \mathbf{R}|), \quad \Delta E_{\text{ref}} = E_{\text{ref}}(t) - E_{\text{ref}}(0). \quad (9)$$

Particles are grouped by component and *initial* radial shell. These measurements compare heating in the initial potential; for live runs they are not changes in the evolving self-potential. Logged total-energy drift is used as a conservation check only in isolation. A prescribed moving nucleus can do work in the tidal calculations.

4 Compact remnants: first-passage response

Both compact distributions retain most of their inner DM. At the common final epoch, $M_{\text{DM}}(< 20 \text{ pc})$ is lower than in isolation by 1.23% for the live cusp and 3.39% for the live core (Table 3). The depletion increases with radius. The isolation runs themselves end only +0.0216% and −0.0308% from their initial 20-pc masses, so the tidal differences are not explained by a comparable secular change in that isolation aperture.

Table 3: Endpoint differences from matched isolation, in per cent, using equation (7). The bound-mass row is an energy proxy, not an exact escaped fraction. The physical time is 88.10216 Myr.

Diagnostic	Cusp live	Cusp frozen	Core live	Core frozen
$M_{\text{DM}}(< 20 \text{ pc})$	−1.23	−1.40	−3.39	−2.66
$M_{\text{DM}}(< 70 \text{ pc})$	−5.35	−5.23	−7.12	−6.35
Bound DM mass proxy	−6.88	−6.82	−8.58	−8.36

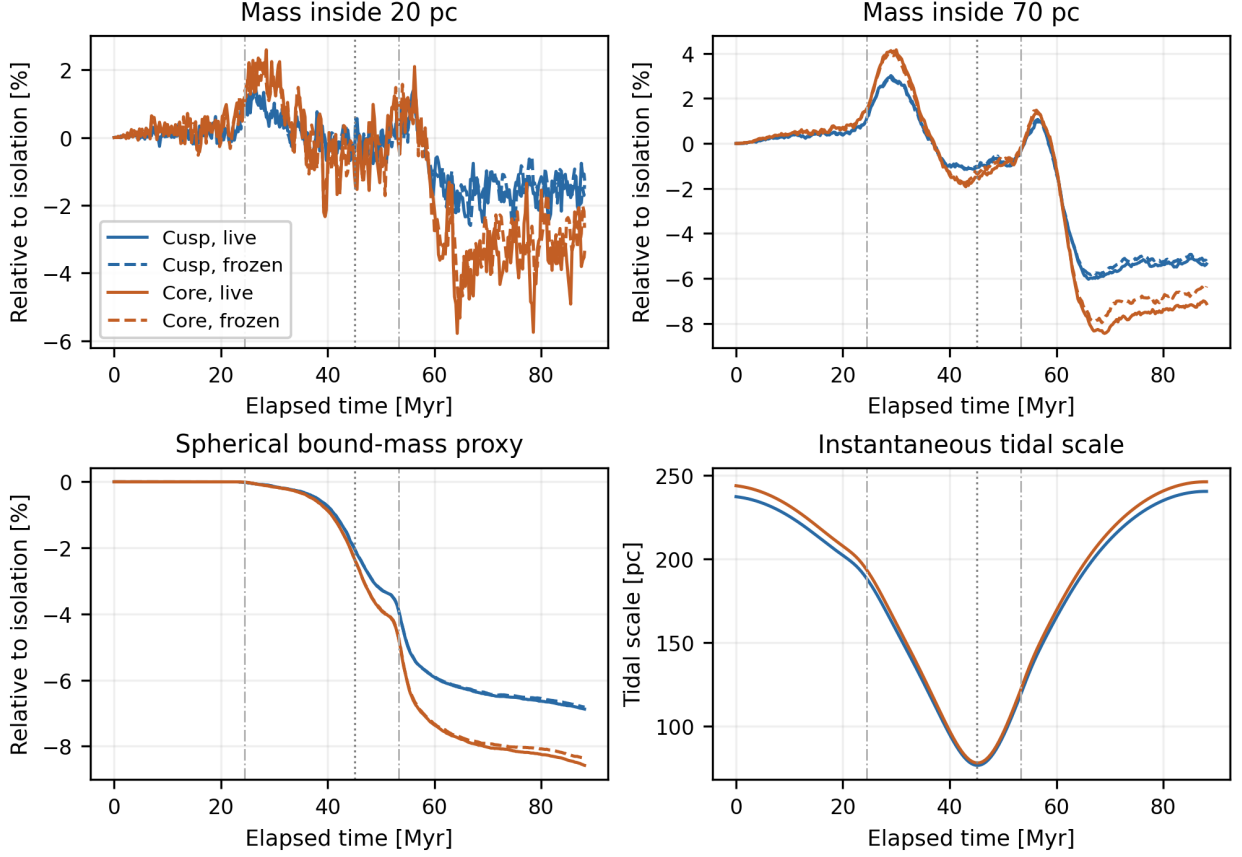


Figure 1: One Galactic passage for the compact cusp (blue) and core (orange). Solid curves show live gravity; dashed curves show frozen potentials. The three mass panels are relative to the corresponding full-interval isolation control. The fourth panel shows the live instantaneous tidal scale. The dotted vertical line marks pericentre; dot-dashed lines mark disc crossings. The largest depletion occurs after the second crossing, while most central DM remains.

The endpoint result persists across the final 10 Myr. The live median 20-pc deficits are 1.41% and 3.31%, with temporal 16th–84th percentile ranges of 1.15–1.96% and 2.43–4.02%.

Frozen medians are 1.43% and 3.03%. Thus the net live/frozen difference is small compared with the common tidal response at this stage. This is consistent with the compact models being dominated centrally by the imposed nucleus. It does not show that frozen gravity remains adequate after repeated stripping or for the much more massive progenitors.

The final-10-Myr shell-density deficits relative to isolation are 2.50% and 4.66% at 20 pc, increasing to about 17% at 70 pc in both profiles. The outer velocity distribution becomes tangential: median $\beta(70 \text{ pc}) = -0.298$ for the cusp and -0.285 for the core, compared with values near zero in isolation. The 20-pc distribution remains close to isotropic. The 70-pc shells retain thousands of effective particles; the 3-pc core shell falls as low as 59. Its large temporal excursions do not establish a resolved central density change.

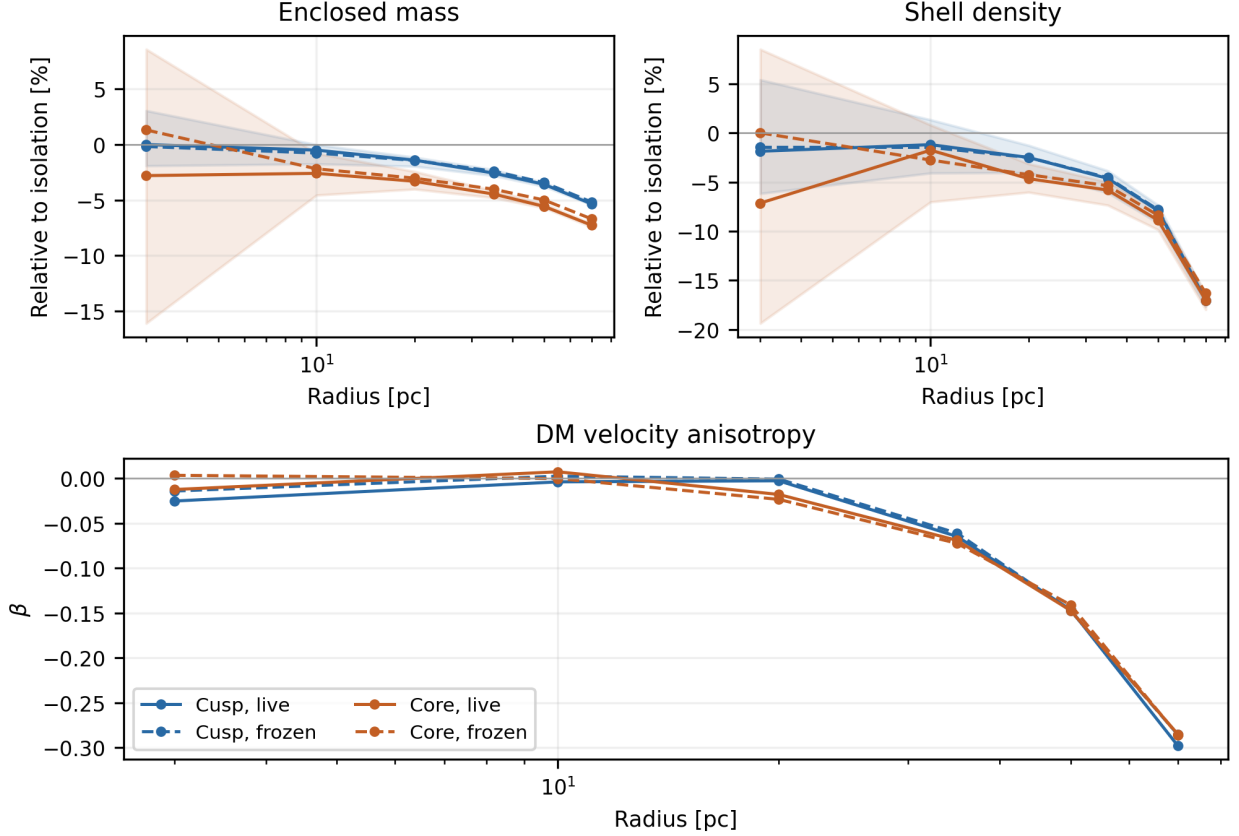


Figure 2: Radial response over the final 10 Myr. The upper panels show median enclosed-mass and shell-density changes relative to isolation; shaded regions are the live temporal 16th–84th percentiles, not confidence intervals. The lower panel shows median DM anisotropy in the tidal runs. Line styles and colours follow Fig. 1. The outer density and velocity distribution change more strongly than the inner mass.

5 Disc crossings and the timing of heating

The Cartesian track crosses the plane twice (Table 4). The second crossing occurs 8.16 Myr after pericentre, at a smaller cylindrical radius than the first. Its strongest compressive host-tensor eigenvalue has magnitude about 0.0323 Myr^{-2} , compared with 0.0169 Myr^{-2} at the first crossing. The tidal scale in equation (8) instead reaches its minimum near pericentre: 76.5 pc for the cusp and 77.9 pc for the core. These quantities describe different parts of the tidal forcing; the minimum tidal scale alone does not capture the compressive disc pulse.

Table 4: Events on the saved orbit. Disc-crossing radii are cylindrical; the pericentre radius is spherical. Times are measured from the initial apocentre. Plane crossings were refined as roots of $z(t) = 0$.

Event	Time [Myr]	Radius [kpc]	v_z [km s^{-1}]
First disc crossing	24.42066	5.09361	−126.98
Pericentre	45.11369	1.99817	—
Second disc crossing	53.27006	3.00873	+175.40

The steepest decline in outer enclosed mass follows the second crossing (Fig. 1). In the frozen cusp, the final median ΔE_{ref} is $8.60 \text{ km}^2 \text{ s}^{-2}$ for particles initially in the 70-pc shell and $0.104 \text{ km}^2 \text{ s}^{-2}$ for the 20-pc cohort. The timing of the compressive pulse, outer heating and subsequent depletion supports a contribution from disc shocking. This batch does not contain a comparison that isolates the disc from the preceding pericentre response, so it cannot assign their individual contributions.

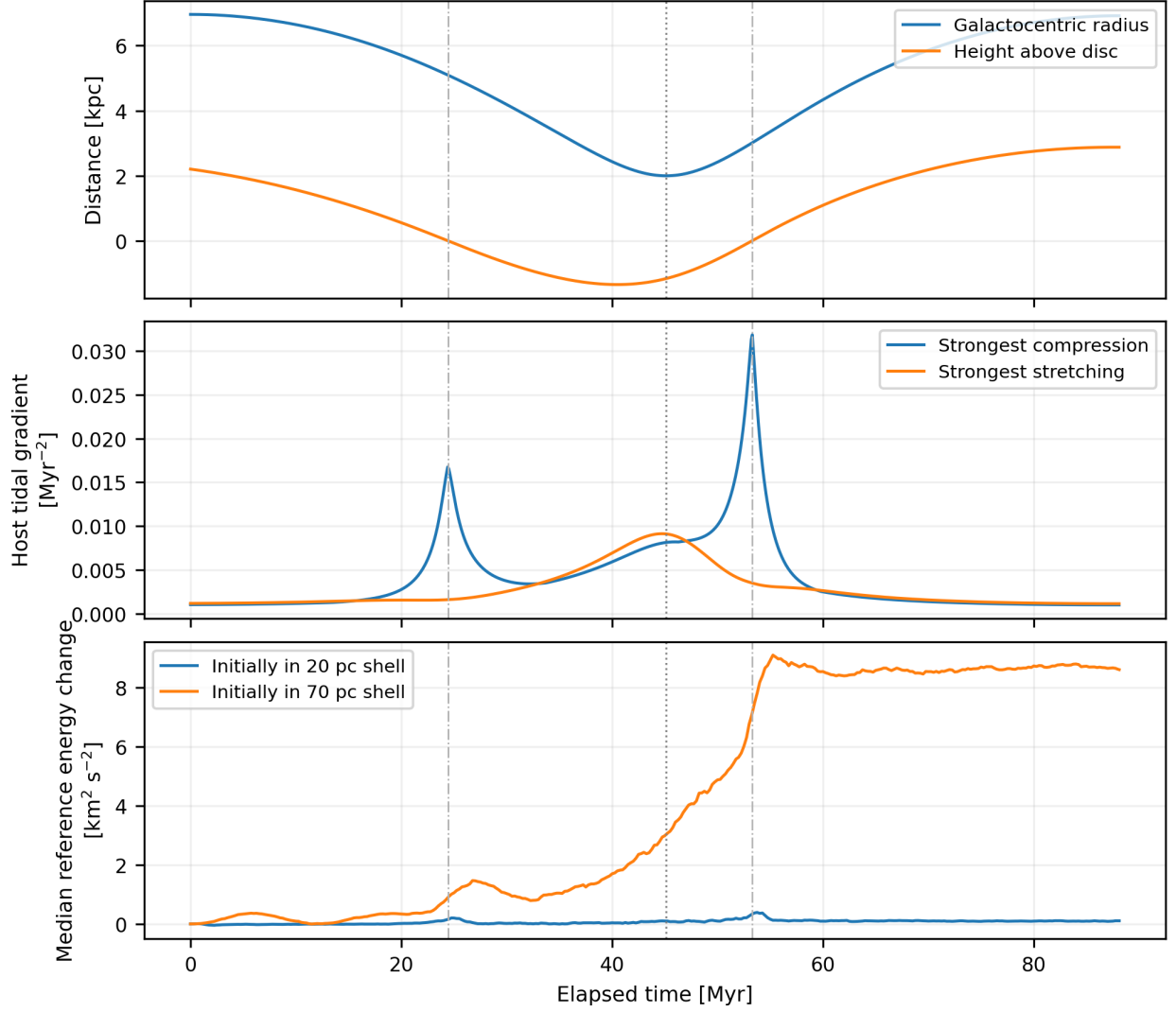


Figure 3: Orbit, host forcing and particle heating. Top: spherical Galactocentric radius and height above the disc. Middle: the magnitudes of the most compressive and most stretching eigenvalues of the inertial host force gradient, without the centrifugal term. Bottom: median changes in the initial-potential energy of frozen-cusp particles selected by their initial 20-pc or 70-pc shells. The outer cohort heats much more strongly. Event markers follow Fig. 1.

6 Do the simulations agree with the analytical predictions?

There is qualitative agreement that the inner remnant changes less than its outskirts, but neither simple analytical prescription gives a quantitative prediction of this first passage. We compare density with density and energy with energy below. An energy increment is not a predicted density decrement, and a one-passage result cannot test a proposed many-Gyr equilibrium.

The earlier estimates in `docs/dm_capture_constraints.tex`, Section 11, and `docs/SECTION11_TESTS.md` used representative pericentres. Here every calculation uses the saved pilot orbit, with $r_p = 1.99817$ kpc and $v_p = 339.99$ km s⁻¹, and the specified initial satellite model. The orbit includes both disc crossings in Table 4. No new evolution was launched for these comparisons.

6.1 Density: instantaneous removal is too strong after one passage

The original Kepler calculation removes all DF contributions with $E > \Phi(r_t)$ from an isotropic, pure $r^{-\gamma}$ tracer around a point mass. For $\gamma > 1/2$ its retained density fraction is

$$F_\gamma(x) = 1 - I_x(\gamma - 1/2, 3/2), \quad x = r/r_t < 1. \quad (10)$$

This is an energy cut, stricter than removing only orbits whose apocentre exceeds r_t . Neither a Plummer-centred potential nor the tapered density in equation (1) satisfies its assumptions. In particular, inserting $\gamma = 0$ into this isotropic Kepler formula is invalid.

We therefore also calculate the energy cut using each actual isotropic initial DF and its combined nucleus-plus-DM potential. With relative potential $\Psi = -\Phi$, binding energy $\mathcal{E} = -E$, and $\mathcal{E}_t = \Psi(r_t)$,

$$\rho_{\text{cut}}(r) = 4\pi \int_0^{\sqrt{2[\Psi(r) - \mathcal{E}_t]}} v^2 f\left(\Psi(r) - \frac{1}{2}v^2\right) dv, \quad r < r_t, \quad (11)$$

and $\rho_{\text{cut}} = 0$ outside. The uncut density uses an upper limit $\sqrt{2\Psi}$. We integrate both over the same finite shells as the simulation diagnostics. The DF is not renormalised and the potential is not updated after removal. Thus this is a specified instantaneous deletion experiment, not a self-consistent equilibrium after stripping.

For a prediction independent of the measured mass loss, we evaluate the tidal scale in equation (8) with the *initial* satellite potential. Its minimum is 76.52 pc for the cusp and 78.01 pc for the core. Figure 4 compares this calculation with the last-10-Myr median shell-density ratios. Table 5 lists representative radii.

Table 5: Retained shell-density fractions. Simulation entries are relative to matched isolation. The analytical energy cut uses the actual initial DF and potential. These entries are density ratios, not enclosed-mass ratios.

Profile	Radius [pc]	Live	Frozen	Energy cut
Cusp	20	0.9750	0.9750	0.8982
Core	20	0.9534	0.9575	0.8529
Cusp	50	0.9208	0.9217	0.5114
Core	50	0.9110	0.9161	0.4926
Cusp	70	0.8294	0.8291	0.1049
Core	70	0.8295	0.8368	0.1190

At 20 pc, instantaneous removal predicts losses of 10.18% and 14.71%, compared with measured live losses of 2.50% and 4.66%. At 70 pc, the predicted losses approach 90%, while the measured losses are about 17%. Using the actual DF improves the comparison with the idealised Kepler cusp, but does not make instantaneous removal a model of the observed response. The Galactic tide varies, the remnant spends a finite time near its minimum tidal scale, and an energy threshold discards some angular-momentum-supported orbits confined within that scale. These differences can account for stronger analytical depletion; their individual contributions have not been isolated. Material outside the instantaneous scale also need not escape immediately. The energy-cut curve is therefore a useful reference for later evolution, not a demonstrated asymptotic state or a rigorous bound on the density after many passages.

6.2 Heating: the scalar pericentre estimate is too small

The existing order-of-magnitude estimate uses

$$\Delta v = T_p r \tau, \quad \tau = r_p / v_p, \quad \omega^2 = GM_n / r^3, \quad A = (1 + \omega^2 \tau^2)^{-\gamma_{\text{ad}}}. \quad (12)$$

For the flattened host we define its scalar radial-force analogue $T_p = -\mathbf{a}_{\text{host}} \cdot \hat{\mathbf{R}} / r_p$. This corresponds to a force-equivalent spherical mass of $1.904 \times 10^{10} M_\odot$, rather than an exact enclosed mass of

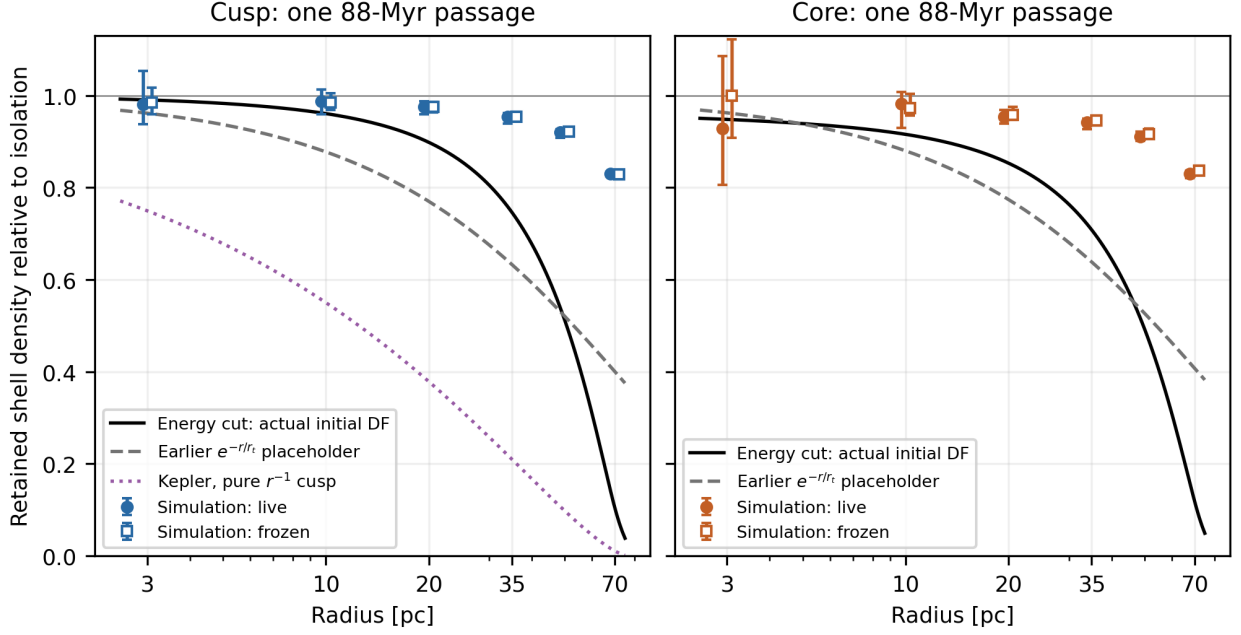


Figure 4: Density predictions and the measured first-passage response. Filled circles and open squares show live and frozen results; bars give temporal 16th–84th percentiles over the final 10 Myr, not independent statistical uncertainties. Points are offset slightly in radius for clarity. The black curve deletes weakly bound phase space from the actual initial DF. The dashed curve is the earlier exponential placeholder; the dotted curve is the idealised Kepler r^{-1} cusp, shown only for the cusp. The latter two are pointwise reference prescriptions. None describes the measured first-passage density profile quantitatively. The core at 3 pc is poorly sampled, with as few as 59 effective shell particles during the passage.

the flattened model. The pilot gives $\tau = 5.75$ Myr. We adopt $\Delta E_{\text{pred}} = \frac{1}{2}(\Delta v)^2 A$ when plotting specific energies. This makes the old dimensionless estimate $(\Delta v)^2 A / (GM_n/r)$ correspond to the characteristic energy $GM_n/(2r)$; it does not equate that scale to every particle’s true binding energy. Geometrical factors of order unity remain approximate.

The power-law adiabatic correction follows the shock calculations of Gnedin, Hernquist & Ostriker and Gnedin & Ostriker [1, 2]. We show $\gamma_{\text{ad}} = 1.5\text{--}2.5$ as a sensitivity range, not a fitted confidence interval. A scalar pericentre duration and frequency do not describe the full host-tensor history, the two disc pulses, or eccentric internal particle orbits.

To measure the response, we retain every particle selected by its initial shell, including subsequent escapers, and compute in the frozen runs

$$\Delta E_i = E_{i,\text{tide}}(t_f) - E_{i,\text{isolation}}(t_f), \quad E_i = \frac{1}{2}|\mathbf{v}_i - \mathbf{V}_n|^2 + \Phi_{\text{initial}}(|\mathbf{x}_i - \mathbf{X}_n|). \quad (13)$$

Frozen potentials provide the cleanest comparison with an imposed-potential estimate. The host can still do work and strip particles. The subtraction removes the matched isolation drift. At 20 pc its mean energy change has magnitude below $2.2 \times 10^{-5} \text{ km}^2 \text{ s}^{-2}$ throughout the last 10 Myr, far below the measured signal. We evaluate the analytical prediction at every selected particle’s initial radius before averaging.

The mean at 20 pc exceeds the corrected prediction by factors of about 4050 and 4400. Even the medians exceed it by factors of 72 and 118. At 70 pc the mean excess is about 17. These discrepancies are much larger than an omitted geometrical factor. Increasing γ_{ad} makes the prediction smaller. Thus the local, single-duration formula is not calibrated by this batch, and fitting an exponent to the discrepancy would mix different physical effects. In particular, the full-orbit mean is not a measurement of heating by the pericentre alone.

The endpoint mean also includes continued host work on escaping particles. Only about 1.1% of either initial 20-pc cohort has positive reference energy at the endpoint, but those particles contribute

Table 6: Frozen full-orbit energy increments and the scalar pericentre prediction with $\gamma_{\text{ad}} = 1.5$. Energies are in $\text{km}^2 \text{s}^{-2}$. Means and medians refer to the same initial-shell cohort. The last column uses its actual mean initial binding energy.

Profile	Radius [pc]	Prediction	Mean	Median	$\langle \Delta E \rangle / \langle E_0 \rangle$
Cusp	20	0.00145	5.88	0.1045	0.0130
Core	20	0.00147	6.46	0.1728	0.0150
Cusp	50	0.402	17.29	2.134	0.0779
Core	50	0.402	19.46	2.251	0.0834
Cusp	70	2.257	37.52	8.602	0.2167
Core	70	2.263	38.64	8.528	0.2102

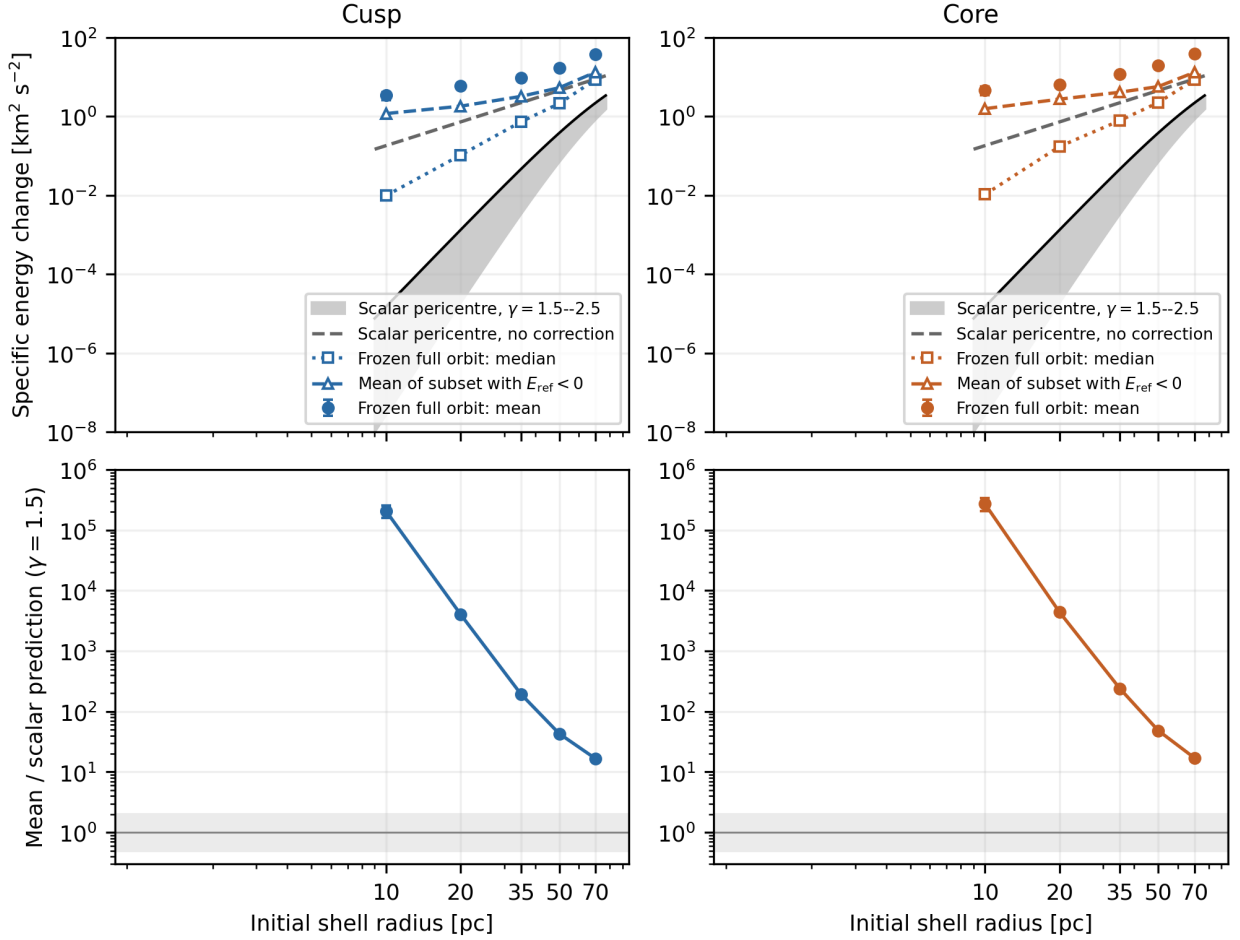


Figure 5: Specific heating versus initial shell radius. Top: the scalar pericentre estimate, with and without adiabatic suppression, compared with the frozen full-orbit mean and median. Triangles restrict the mean to particles with negative endpoint reference energy. Error bars on the mean are conditional particle-sampling standard errors; they exclude modelling and convergence uncertainty. Bottom: the mean divided by the shell-averaged prediction at $\gamma_{\text{ad}} = 1.5$; the grey band denotes agreement within a factor two. The 3-pc shell is omitted because it lies inside the Plummer core, where the assumed Kepler frequency is inappropriate. This figure tests the adequacy of the scalar prescription for the full passage; it does not isolate a pericentre shock from disc forcing or calibrate an adiabatic index.

69% of the cusp’s net energy increment and 58% of the core’s. Restricting the measurement to negative reference energy gives mean increments of 1.82 and $2.75 \text{ km}^2 \text{ s}^{-2}$, still far above the local prediction. This is a selection diagnostic, not an exact bound/unbound classification in the Galactic field. An endpoint cohort mean should not be interpreted as heat retained by a closed, bound subsystem.

The large ratio to a very small analytical prediction does not imply destruction of the inner remnant. The 20-pc cohort’s mean energy increment is only 1.30% or 1.50% of its mean initial binding energy, while its median increment is much smaller still. At 70 pc the corresponding mean fractions are 21.7% and 21.0%. These energy fractions belong to fixed particle cohorts; the density measurements count whichever particles occupy a shell at the later epoch.

6.3 Why a particle at 20 pc is not a 20-pc orbit

We compute each particle’s apocentre in the initial spherical potential, before applying the host. Among particles initially in the 20-pc shell, 28.7% of cusp particles and 38.2% of core particles have apocentres beyond 40 pc. Fractions of 7.1% and 9.5% reach beyond the minimum tidal scale. Their larger excursions expose them to stronger differential forces than a calculation that keeps them at 20 pc.

Splitting the initial cohort at an apocentre of 40 pc makes this explicit. Over the last 10 Myr, the mean increments for confined orbits are 0.083 and $0.015 \text{ km}^2 \text{ s}^{-2}$ for cusp and core. The means for the more extended orbits are 18.38 and $15.74 \text{ km}^2 \text{ s}^{-2}$. Weighted by their cohort fractions, these extended orbits supply 99–100% of the mean increment. They also explain why the mean greatly exceeds the median. This comparison averages increments per particle over saved outputs, so its values differ slightly from the endpoint entries in Table 6.

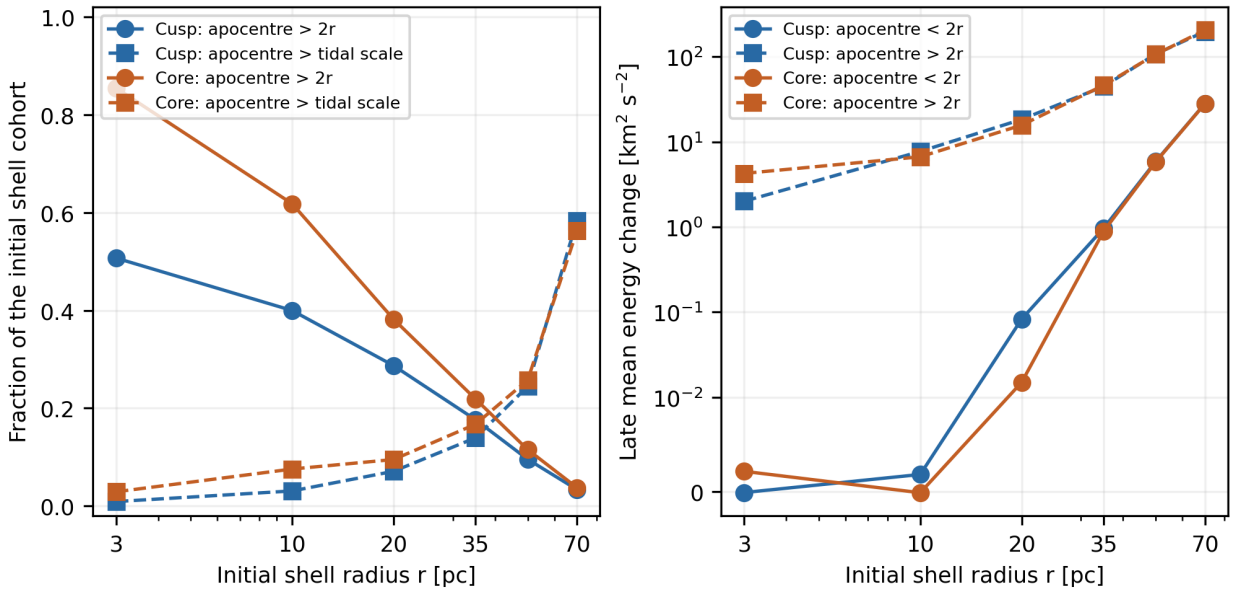


Figure 6: The initial shell radius does not fix an orbit’s radial extent. Left: fractions of each initial cohort with apocentres beyond twice the shell radius or beyond the minimum tidal scale. Right: mean frozen tidal-minus-isolation energy increments, averaged over the final 10 Myr, split by initial apocentre. Extended orbits dominate the mean heating of the inner cohorts. The right axis is linear very close to zero and logarithmic away from zero. Lines connect the measured cohorts.

The observed hierarchy of inner and outer response therefore agrees with the broad physical expectation. Quantitative agreement requires a prediction that follows the actual internal orbits and time-dependent tidal tensor, including the disc crossings and delayed escape. The present comparison does not determine a unique disc contribution, an asymptotic retained density, or a lifetime survival probability. Those remain questions for the repeated-passage experiments and

their numerical controls. The physical stellar-heating timescale is also a separate prediction: these collisionless runs do not simulate diffusion by real stars and cannot validate that rate.

7 Numerical sensitivity and the force-kernel repair

The 20.00762 Myr isolation controls vary timestep, softening and particle number separately. The first two comparisons retain the same initial phase space and masses. Their endpoint $M_{\text{DM}}(< 20 \text{ pc})$ differences are at most 0.116% (Table 7). The doubled- N runs use different samples; their own live/frozen differences are +0.011% for the cusp and -0.155% for the core. Their offsets from the baseline should therefore not be interpreted as pure numerical bias. All these isolation runs retain their initial bound DM mass proxy.

Table 7: Short-control endpoint $M_{\text{DM}}(< 20 \text{ pc})$ differences from the original live baseline, and maximum logged energy drift. Energy entries are $100 \max_t |[E(t) - E(0)]/E(0)|$. The half-softening cusp uses the repaired force kernel.

Control	Mass difference [%]		Energy drift [%]	
	Cusp	Core	Cusp	Core
Baseline	—	—	0.02930	0.01677
Half timestep	+0.116	−0.062	0.00617	0.00266
Half softening	+0.091	+0.031	0.01571	0.00696
Double particle number	−0.642	−0.046	0.02942	0.01652

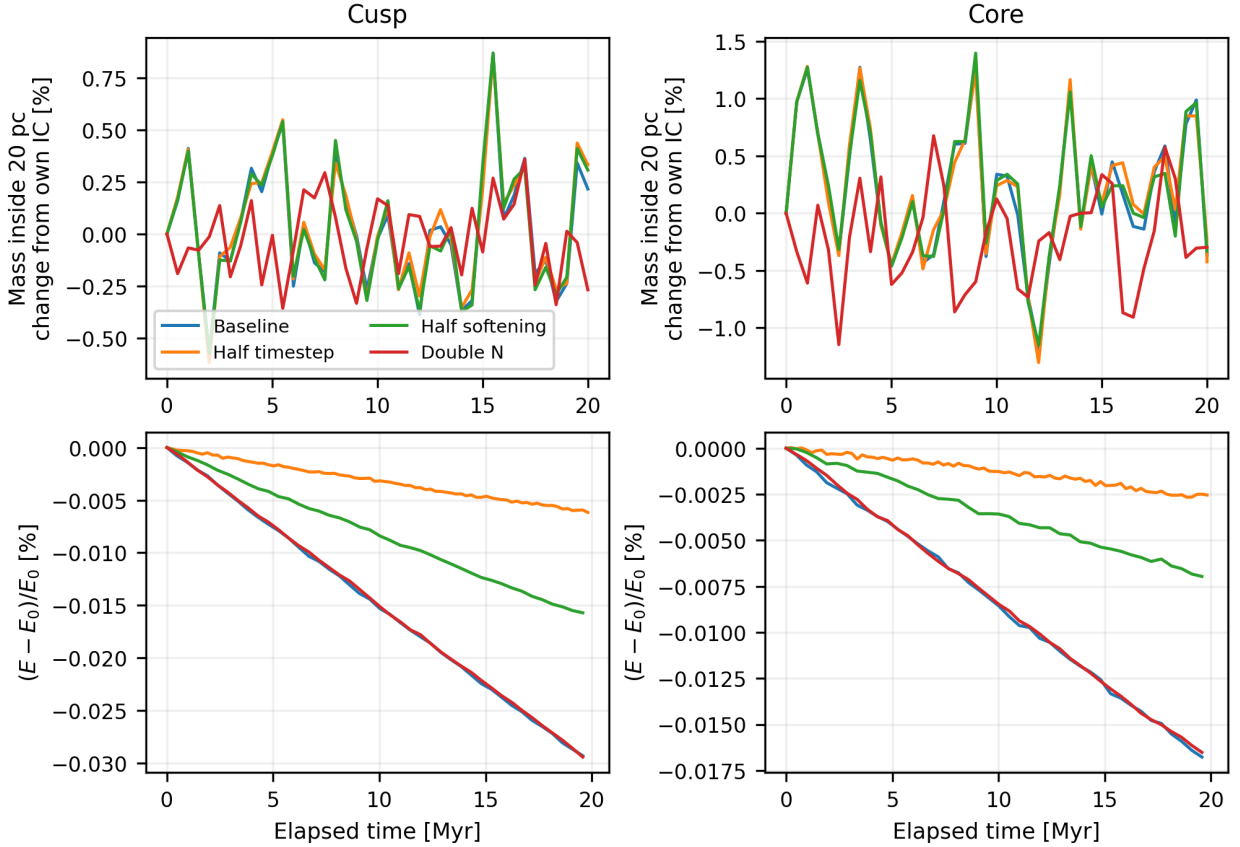


Figure 7: Remnant isolation sensitivity over 20 Myr. Upper panels show $M_{\text{DM}}(< 20 \text{ pc})$ relative to each run’s own initial sample, so they need not end at the baseline differences in Table 7. Lower panels show logged energy changes; the same colour mapping applies in all panels. Halving the timestep reduces the secular energy drift. The different doubled- N mass fluctuations include a change in particle realisation.

The longer baseline isolation runs reach maximum energy drifts of 0.135% and 0.0728% over 88 Myr. Although their endpoint inner masses remain stable, the secular trend favours the finer timestep for longer experiments. No linear extrapolation of that drift to a many-Gyr integration is assumed. The short controls support the pilot interpretation, but they have not tested resolution sensitivity under the full 88-Myr tidal forcing. There is also only one sampling seed.

The initial cusp run with 0.1 pc softening failed because an intermediate coefficient in the installed single-precision P1 Taylor kernel overflowed. An unchanged repeat reproduced the failure.

With $Q = r^2 + \epsilon^2$ and $H = \epsilon^2/2$, the repaired implementation evaluates the same coefficients as

$$D_n = T_n \left[1 + (2n + 1) \frac{H}{Q} \right], \quad T_n = M(2n - 1)!! Q^{-n-1/2}, \quad (14)$$

without forming an unnecessary higher derivative that overflows before being multiplied by H . Double-precision intermediates are used; native coefficients that genuinely exceed range are rejected explicitly. The repair preserves the force law, masses, softening and timesteps. A run-local library replaces only the kernel object; the installed NEMO source and library are unchanged.

The original 10^5 -particle configuration then completed the full 20.00762 Myr, giving 41 analysed snapshots. A 64-particle close-cell reproducer fails with the original library and passes with the repair, with maximum force error 7.25×10^{-5} relative to direct P1 summation. The focused suite passed 95 tests, followed by five kernel/integrity tests after the runtime checks were strengthened. The original failures and instrumentation attempt remain preserved. Other completed batch runs used the original kernel; the repaired kernel has not yet been used to repeat the full tidal passage.

8 Extended progenitors in isolation

Both progenitor pairs complete 100.00826 Myr. Their DM and stellar bound-mass proxies remain constant, with maximum logged live energy drifts of 0.00244% for the cusp and 0.00114% for the core. At every saved epoch, the fraction of mass inside the 100-pc protection radius contributed by particles heavier than ten times their component’s minimum mass is zero. This is the defined heavy-particle diagnostic; it does not exclude all effects of the mass spectrum or finite-particle relaxation.

8.1 Mass profiles on the scale of the extended body

The original diagnostic apertures stop at 70 pc. We therefore independently read the final snapshots and extend the enclosed-mass measurements to 100, 200, 500, 1000, 2000, 5000 and 10000 pc. At every one of these radii, the live and frozen endpoint masses agree within 1% for both components in both progenitors. At 1 kpc, the DM differences are -0.38% and -0.53% , while stellar differences are -0.35% and -0.14% for cusp and core, respectively.

These endpoint comparisons support using the models for first experiments on disruption of the extended body. They do not provide a complete time series at those larger radii; the next runs should save diagnostics there at every output time. The live/frozen agreement is a pilot stability check, not a convergence demonstration across particle numbers or refinement schemes.

8.2 Central particle counts and their consequences

Central resolution differs strongly between profiles (Table 8). The analytic cored DM mass inside 20 pc is only $2772M_\odot$, compared with $149950M_\odot$ for the cusp. The initial core sample contains 16 particles there, corresponding to $2142M_\odot$; it was already below the continuum target before evolution.

The core ends with 18 live and 23 frozen DM particles inside 20 pc. Expressed relative to its initial sample, these become apparent mass increases of 12.5% and 43.75%. Both series fluctuate

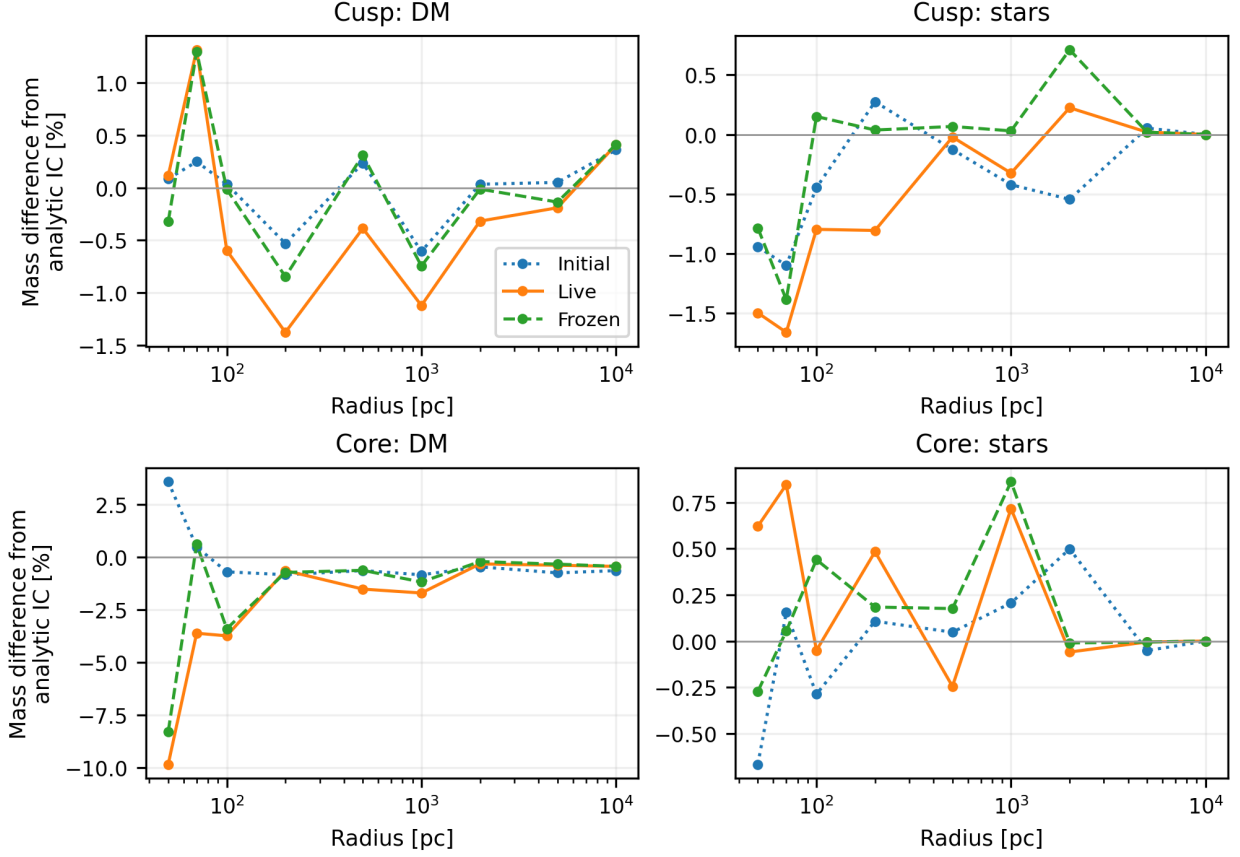


Figure 8: Progenitor enclosed masses relative to the analytic continuum IC profile. Dotted lines show the initial particle sample; solid and dashed lines show live and frozen endpoints at 100 Myr. Both DM and stars remain close on the scales of the extended body. The larger cored-DM fluctuations at 50–100 pc must be read with the smaller effective counts. The plotted offsets from the analytic model include initial sampling fluctuations.

Table 8: Initial central resolution of the progenitor models. DM particle weights inside these apertures are equal, so the listed aperture effective counts coincide with particle counts. Shells have width ± 0.1 in $\ln r$.

Quantity	Cusp	Core
Analytic $M_{\text{DM}}(< 20 \text{ pc})$ [$M_{\odot}$]	149950	2772
Sampled $M_{\text{DM}}(< 20 \text{ pc})$ [$M_{\odot}$]	152315	2142
DM N_{eff} inside 20 pc	539	16
DM N_{eff} in the 20-pc shell	205	15
Stellar N_{eff} inside 20 pc	669	692

strongly and both have a median aperture count of 20. These percentages do not establish physical central growth. Its shell density and anisotropy are also poorly sampled. The cusp’s several hundred central DM particles can reveal gross changes, but small shell-density differences remain noisy.

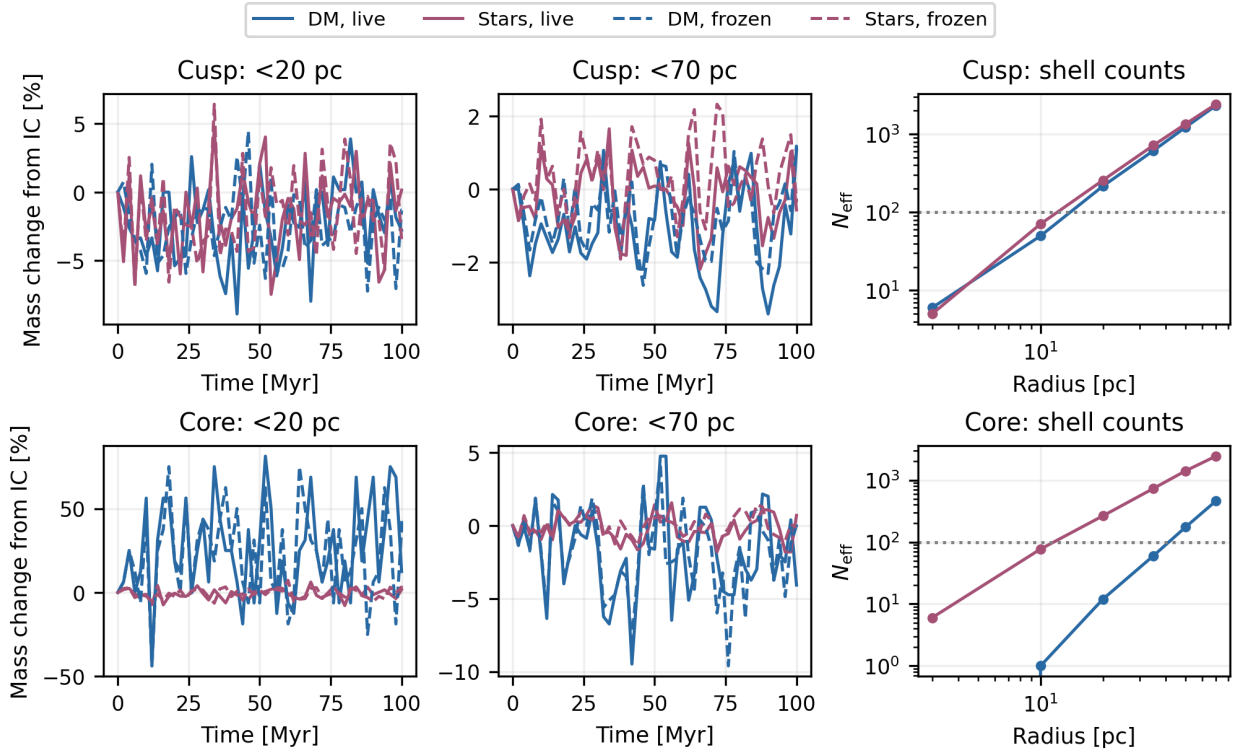


Figure 9: Progenitor isolation time series and effective shell counts. Blue curves represent DM and purple curves stars; solid and dashed lines denote live and frozen evolution. Left and middle columns show enclosed-mass changes from each initial sample. The right column shows median live shell counts; the dotted $N_{\text{eff}} = 100$ line is a visual reference, not a convergence criterion. The cored halo’s large central fluctuations occur in both engines and coincide with very small DM counts.

9 Density changes at 20 pc across the batch

Figure 10 collects all 24 completed evolutions in the 18.1–22.1 pc shell, a local density diagnostic distinct from $M_{\text{DM}}(< 20 \text{ pc})$. Tidal runs are compared with matched isolation at the same time; isolation runs are compared with their own initial samples.

The median tidal shell-density changes over the final 10 Myr are -2.50% for both cusp engines, -4.66% for the live core and -4.25% for the frozen core. Doubled- N remnants in isolation show median increases of $0.35\text{--}0.77\%$ from their own initial densities. Their different particle samples prevent attributing this sign change alone to particle number.

The progenitor cusp has median increases of 3.90% and 6.59% in its live and frozen runs. The core’s larger excursions remain unresolved with only 15 initial effective shell particles. These results describe the completed pilot configurations, including one tidal orbit.

DM shell density at 20 pc: all 24 completed evolutions

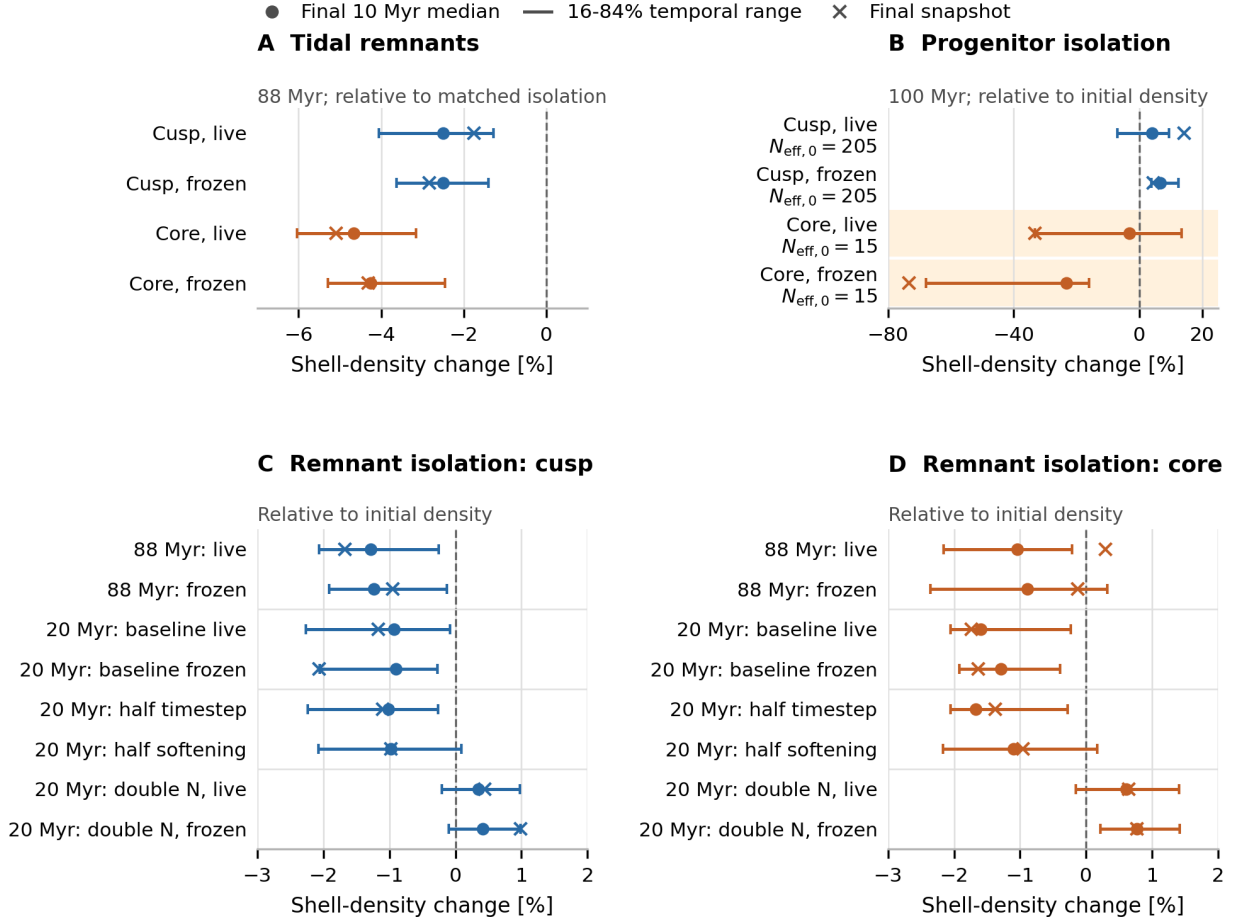


Figure 10: DM shell-density changes at 20 pc across the full batch. Blue denotes cusps and orange cores. Panel A shows the tidal effect relative to matched isolation after one Galactic passage. Panels B–D show changes from each run’s own initial shell density; their durations are labelled. Circles give medians over the final 10 Myr, bars the temporal 16th–84th percentiles, and crosses the final snapshot. The horizontal scales differ. Shaded progenitor-core rows are unresolved at this radius; the labels give initial effective shell counts. The lower panels include every completed remnant isolation control, including the repaired half-softening cusp run.

10 What the batch supports and what comes next

For the adopted compact models, one passage removes relatively little inner DM while changing the outer density and anisotropy more strongly. The cored distribution responds more strongly under the chosen inner-mass matching convention. Disc-crossing timing offers a specific physical hypothesis for part of the response. These results justify pursuing repeated-passage survival, once numerical sensitivity has been checked under the full forcing.

The next remnant comparison should extend the timestep and particle-number controls to the entire tidal interval, with corresponding isolation and frozen controls and the repaired kernel. A second seed would measure realisation variation that the temporal bands cannot. The 20- and 70-pc responses should be assessed before expanding the physical grid. A later comparison of host components could isolate the disc’s contribution.

The progenitor programme can proceed independently towards an extended-body disruption pilot on a specified orbit. Central-remnant measurements need better sampling first, especially for the core. Simply doubling its present particle number would leave only a few tens of DM particles

inside 20 pc; the sampling design requires attention, followed by its own isolation check. Diagnostics over 0.1–10 kpc at all output times should accompany the next tidal runs.

All conclusions remain conditional on the imposed nucleus, orbit and initial models. A self-consistent historical reconstruction would additionally need the coupling of mass loss to orbital evolution and a responsive nucleus. The current batch contains no progenitor tidal evolution and no repeated remnant passages. No new simulation was launched to prepare this report.

A Run provenance and reproduction

The analysis comprises 24 completed evolutions from 14 preparations. It excludes the original failed softening run, its unchanged repeat and its instrumented failure. The successful replacement is `remnant_cusp_eps_half/evolution_live_p1_fixed`.

The audit verified prepared inputs, all 24 diagnostic checksums, manifest links, durations, monotonic times, mass invariance, nested apertures and matched initial conditions and host/orbit files. Twelve independently read final snapshots passed ID, mass and epoch checks and reproduced the saved aperture masses. The expanded profile table has 264 rows; the analysis records 279 input hashes, including the snapshot files used in these measurements.

The principal project-relative locations are:

- `results/dynamics/isolation_20260921T113007Z/`: baseline remnant and progenitor isolation runs.
- `results/dynamics/remnant_passage_20260921T114939Z/`: tidal runs, full-interval isolation references and numerical controls.
- `results/repairs/numerics_20260921T125417Z/`: force-kernel repair evidence, tests and successful replacement record.
- `results/dynamics/batch_analysis_20260921_v2/`: numerical summary, endpoint profiles, orbit context, analysis source copy and checksums.

The calculation is reproduced from the project root with `bin/analyse_dynamical_batch.py`, using a fresh output directory:

```
export PYTHONPATH=src OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1
python bin/analyse_dynamical_batch.py \
--output results/dynamics/batch_analysis_repeat
```

The pipeline and companion analysis are documented in `docs/DYNAMICAL_EXPERIMENTS.md` and `docs/DYNAMICAL_BATCH_ANALYSIS.md`.

Figure scripts are `bin/plot_dynamical_writeup.py`, `bin/plot_density_changes_20pc.py` and `bin/plot_density_changes_20pc_paper.py`. JSON records in `results/plot_data/` give their values and checksums; all required figures and provenance are included in `output/source/dynamical_batch_latex.zip`. The analytical comparisons are reproduced with `bin/compare_dynamical_analytics.py`. Their three PNG figures and `results/plot_data/dynamical_analytics.json` record the numerical values, definitions and input hashes. The independent signed-DF inversion recovers the input density to better than 2×10^{-7} over the tested 2.2–85 pc range. Doubling its energy grid and quadrature changes the shell-retention fractions by less than 1.1×10^{-7} . Three independent tests check the Kepler energy-cut limits and finite-shell integration. Particle IDs, masses, matched times and audited snapshot hashes are checked before the energy comparison; all existing simulation products are preserved. Compile from `docs/` with `latexmk -pdf dynamical_batch.tex`.

References

- [1] Gnedin, O. Y., Hernquist, L., & Ostriker, J. P. 1999, *Tidal Shocking by Extended Mass Distributions*, ApJ, 514, 109–118; doi:10.1086/306910; arXiv:astro-ph/9709161.
- [2] Gnedin, O. Y., & Ostriker, J. P. 1999, *On the Self-consistent Response of Stellar Systems to Gravitational Shocks*, ApJ, 513, 626–637; doi:10.1086/306864; arXiv:astro-ph/9902326.